Bachelor of Science (BS1): ESGIS PRELIMINARY LINGUISTIC PERFORMANCE REVISIONS

TASK ONE: Match the sentences or utterances under column A with the most appropriate words or phrases under column B

COLUMN A	COLUMN B
a) How are you feeling now?	1) Location.
b) How are you doing?	2) Identification.
c) Where do you come from?	3) Schedule/timetable
d) What is this?	4) Recovery/ requesting health details
e) How can I address you?	5) Greeting
f) What are you reading?	6) Field of study
g) Who is she?	7) Date
h) Would you drink a cup of tea?	8) Offer/ implied invitation
i) When is the English class?	9) Introduction
j) What time is the train coming?	10) Naming/christening.

TASK TWO: PRONUNCIATION TIPS

A. Revising the alphabet and making difference between graphemes and phonemes.

B. The suffixed "ed or d" added to form the preterit or past participle of regular verbs or even adjectives are pronounced /t/, /d/ or / Id/. Read the following terms and classify them under three columns entitled /t/, /d/ or /Id/.

Word list

Planted, played, pleaded, pleased, pledged, plotted, plucked, called, phoned, dated, denoted, denounced, denuded, depended, laughed, deployed, employed, depraved, corrupted, depressed, depreciated, deprived, deputized, derided, ridiculed, mocked, derived, descended, described, desecrated, selected, extracted, excepted, deserted, abandoned, quitted, detached, detected, developed, founded, connected, fired, transferred, transported, pounded, celebrated, blinded, mended, tested, minded, handed, blended, fused, evoked, wondered, wandered, wanted, demanded, slapped, decided, commanded, shared, checked, chopped, possessed, ploughed, mixed, introduced, pretended, guarded, visited, launched, engaged, involved, locked, scored, conversed, converted, addressed, added, convicted, convoked, provoked, cooked, copied, approached, ended, expected, parked, emphasized, designed, designated, associated, learned, jumped, leaped, worked, planned, plagiarized, regained, pleaded, deterred, determined, capsized, tended, attempted, tempted, terrified, thanked, changed, accused, charged, equated, evaded, avoided, evaluated.

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TASK THREE: GRAMMAR REVISIONS

A. TURN THESE SENTENCES INTO NEGATIVE AND INTERROGATIVE FORMS

1. John put two letters in the box of the head of studies.
2. You've got the news from the headmaster.
3. He's playing with the tail of the lion.
4. She's been managing this firm for many years.
5. This man's used to having his dinner in that restaurant.
6. Blade used to cheat in class in his childhood.
7. The shareholders let the investors make the decisions.
8. You'd better divorce your husband for a much better life in America.
9. You'd rather read your lessons than watch TV.
10. I'd probably say a few words to this delinquent guy.
11. You'd got your bachelor's degree before getting your master's degree.
12. The lad has a lot of money in his pocket.
13. Women have to participate in the sovereignty expense.
14. I need to play my trump card.
15. They need make a right decision about their wedding.

B. WRITE THE BRACKETED VERBS IN THEIR CORRECT TENSE AND FORM. DON'T COPY THE WHOLE SENTENCE.

1. Ahammouto (**to drink**) palm wine for three hours now but has not got drunk yet.
2. No sooner the announcement (**to be**) advertised on the radio than thousands of applications (**to flow**) into the manager's office.
3. I am fed up with this situation! I will stop (**to work**) for this ungrateful firm!
4. Computers (**not to make**) in Cotonou.
5. It is about two years that they (**to ban**) the use of nylon on the campus.
6. Everybody (**to wait**) for the results, aren't they?
7. That lady (**to be married**) to Sarkozy for teh years before she (**to divorce**) a few years ago.
8. When the trial started, Gbagbo and Blégoudé realized that they should (**to think**) hard before sticking to power and killing thousands of civilians after the outbreak of the war.
9. Hardly Koinnon (**to write**) his exam when he (**to call**) his parents to let them know that he did not do well.
10. If the national Hagbé (**to win**) the 2011 presidential election, he (**can sell**) this country to any buyer because of his obsession for cash.

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C. CHOOSE THE MOST APPROPRIATE OPTION FROM THE BRACKETS. WRITE DOWN JUST THE NUMBER AND YOUR SELECTION

1. On the day (**when; where**) that nice-looking lady accepted my invitation I thought I was in the seventh heaven.
2. The sports hall is (**when; where**) the students do the examination.
3. I'd rather you (**learn; learnt; to learn**) your lessons today evening.
4. I would rather they (**will call; called**) me tomorrow since I will be free at this moment.
5. I would rather you (**brought; had brought**) the bag yesterday.
6. What would I do if I (**was; were**) (**at; in**) your place? Ok, what about dinner (**at; in**) my place?
7. (**Are; do**) you agree that he uses your finance textbook?
8. (**How; what**) do you call it? I think it is garlic.
9. The wife of Mr. Achènon, Mrs Folinnon did nothing all the daylight except (**watching; watch**).
10. John sent (**a; an**) SMS to Bella who is in (**a; an**) universe where there is no network.
11. The teacher stops (**to pick up; picking up**) the cell phone when he gets an income call.
12. This carrier requires that he (**signs; sign**) the delivery note as a proof of reception.
13. The man (**who/ whom**) you were talking to is the one (**who/whom**) I gave the bag to?
14. Why are you apologizing? It is john and (**I; me**) who must apologize lady.
15. You can imagine that my sugar mummy and (**I; me**) were among the special guest.
16. (**Who; Whom**) does John believe is the whore among these young ladies?
17. Paul was one of my classmates so you can guest that he is a friend of (**me, mine**).
18. If everybody had come to the meeting with (**their; his**) report, the meeting would have gone a lot more smoothly.

D. ASK A QUESTION BASED ON EACH SENTENCE; START WITH THE BRACKETED QUESTIONS.

1. The government sent military force to Nigeria (**Whom…. Where**)
2. You would not have agreed on the deal if you had informed me. (**What….**)
3. When he was twenty-seven he got his first daughter. (**How old…**)
4. This country is too cold and too hot. (**What …**)
5. I was not a teacher in 2012. (**What…**)

E. ASK QUESTIONS ON THE UNDERLINED WORDS IN EACH SENTENCE

1. They have fetched <u>a lot of</u> water.
2. I need John <u>to send this letter through the post office</u>
3. The new wedded were <u>very attractive</u> on this picture than the others
4. It took me <u>two years</u> to complete my master's degree.
5. This young boy <u>regularly</u> pays visit to his sugar mummy.

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A. Vowel Sounds

There are twelve monophthongs or mono-vowels and eight diphthongs or double vowels in the English language.

✓ **Mono-vowels or monophthongs**

Vs 1 [i:] : fleet, feat, greet, flee, heal, heel, lead, bee, be, eat, team, piece, peace
VS 2 [ɪ]: hit, bit, mid, middle,
VS 3 [e]: bed, beg, level, egg, elephant,
VS 4 [æ]: bag, bad, cat, mat, mas,
VS 5 [ɑ:]: car, bar, father,
VS 6: [ɒ]: lot, lottery, pot,
VS 7 [ɔ:]: four, for, fore, floor, paw,
VS 8 [ʊ]: book, good, look, wood, put, could, would,
VS 9 [u:]: zoo, you, who, too, two, to,
VS 10 [ʌ]: flood, blood, but, love, bus, enough,
VS 11 [ɜ:]: sir, bird, further, girl, whirlwind, nurse, turn,
VS: 12 [ə]: later, letter, error, professor,

Recap: The twelve pure vowels are / i:, ɪ, e, æ, ɑ:, ɒ, ɔ:, ʊ, u:, ʌ, ɜ:, ə/). Pure vowels are subdivided into short vowels i.e /æ, e, ɪ, ɒ, ə, ʊ, ʌ/ and long vowels /ɑ:, i:, ɜ:, u:, ɔ:/).

- **Double vowels or diphthongs**

There are eight diphthongs which are all tense vowels or long vowel:
/ɔɪ, eɪ, aɪ, ɪə, eə, ʊə, aʊ, əʊ,/.

✓ **Front diphthongs /ɔɪ, aɪ, eɪ/**

/ɔɪ/ e.g. boy, toy, employ, deploy, boil, toil, choice, joy, rejoice
/aɪ/ e.g. sky, I life, fine, shine, bind, tie, lie, fight, night, price, try, shy, buy
/eɪ/ e.g. Monday, today, lay, clay, late, blame, plane, claim, explain, eight, veil

✓ **Back diphthongs /əʊ, aʊ/**

/əʊ/ e.g. go, tomato, no, approach, coach, toad, toe, hoe, cold, scold, know, low,
/aʊ/ e.g. how, now, bow, mouth, aloud, sound

✓ **Central diphthongs /eə, ʊə, ɪə/**

/eə/ e.g. bare, share, dare, air, stair, vary, Mary, there, their, wear, pear, bear,
/ɪə/ e.g. beer, sheer, clear, dear, beard, here, mere, idea, area,
/ʊə/ e.g. tour, cure, pure, sure, mature,
Note: There are also triphthongs which will be discussed later on.

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B. Consonant Sounds

There are 24 consonant sounds in the English language. English consonant sounds are either voiced or voiceless. Here, all the emboldened ones are voiced.

STOP

[p] potato, pig, pink, plate, plane pallor
[b] banana, bird, bed, bedroom, black
[t] television, table, teacher, tea, tree, tiger,
[d] dad, dance, daytime, daybreak, dawn, duck, dromedary, dog,
[k] cabbage, couch,, curtain, cat, camel
[g] guava, gadget, grenade, game, goal, green, gift, gadget, goat, goose

FRICATIVE

[f] fan, factory, fabric, fairy, February
[v] vacation, vaccine, vegetable, vicar, villa, village, vinegar, viper
[θ] thought, thin,
[ð] this, that, the, there, then
[s] serpent, soup, sauce, sugar, September, sun, sand, south
[z] zap, zeal, zapper
[ʃ] shop, shade, sheep,
[ʒ] pleasure, leisure; measure,
[h] helicopter, honey, hare,

AFFRICATE

[tʃ] channel, chain, children, church, chicken, chick
[dʒ] January, June, July, jury, judge, juice, junction

FLAP

[ɾ] letter, pity, matter,

NASAL

[m] money, monogamy, Monsignor, moon, midday
[n] north, neck, nectar, needle, nee, nation, nausea, nest, next, net, new, noon
[ŋ] singer, something, language, ring,

APPROXIMANT

[r] red, rat, rubbish, rubber,

LATERAL

[l] lake, lagoon,

GLIDE

[w] win, winner, white,
[j] yet, year, yes, yeast, yellow

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Recapitulation

Voiceless consonants	Voiced consonants
[p]	[b]
[t]	[d]
[k]	[g]
[f]	[v]
[θ]	[ð]
[s]	[z]
[ʃ]	[ʒ]
[h]	
[tʃ]	[dʒ]
	[m]
	[n]
	[ŋ]
	[r]
	[l]
	[w]
	[j]

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01

INTRODUCTION TO COMPUTING

INTRODUCTION TO COMPUTING

1.1 What is Computer?

Def1: An electronic machine that can store and process information. It is defined as stored program digital computing system.

Def2: A Computer is an electronic device of wires, transistors, circuits, instructions, and data can transmit, store and manipulate information.

Def3: A Computer is a machine that accepts data and processes that data (data may be numbers, letters or both or even sounds). Information is turned into electrical pulses so that it may be processed by sorting, collating and deleting mathematical manipulation and other forms of data processing.

Def4: A computer accepts information using an input device. The information is processed by a central processing unit or stored in a storage unit i.e. Hard Disk and then processed.

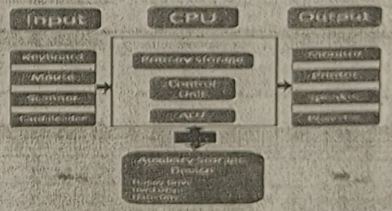

Figure 1: A block Diagram of Computer System

The result is supplied by output devices. If the input and output units are connected to a manufacturing process, the computer can control the process and the system is called a computing system.

1.2 Basic Computer Operations

To understand computer basic concept one must know about the 4 functions or Operations of the computer.

1.2.1 Input Operation

The computer receives information from users. A user enters information using input devices like keyboard, mouse or any other

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devices like webcam, punch card, magnetic tapes, joystick or magnetic disk etc. The Input unit accepts information using input devices then it converts the given data to readable form and this data moves to Central Processing Unit (CPU).

1.2.2 Storage Operation

The given information stored in a computer using the different storage devices i.e. central process unit and auxiliary memory. The auxiliary memory is also known as secondary or external storage have hard devices for example Floppy, Hard Disk, Compact Disk and Flash Drive. These different storage devices have both advantages and disadvantages. Auxiliary storage speeds up information and stores it long term and permanent.

1.2.3 Processing Operation

It is considered the basic computing operation. It executes the instructions, control storage data and input or output devices attached the computer.

1.2.4 Output Operation

Last but not the least the output unit which represents results from the operations of central processing unit CPU. The result may be in hard or soft form i.e. visual display unit, printers, and headphones.

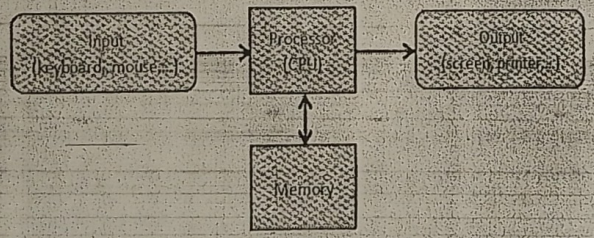

Figure 2: A Diagrammatic Representation of Computer System

1.3 History of Computer

The computer as we know it today had its beginning with a 19th-century English mathematics professor named Charles Babbage. He designed the Analytical Engine and it was this design that the basic framework of the computers of today are based on. Generally

speaking, computers can be classified into three generations. Each generation lasted for a certain period of time, and each gave us either a new and improved computer or an improvement to the existing computer.

1.3.1 First Generation 1937 – 1946

In 1937 the first electronic digital computer was built by Dr. John V. Atanasoff and Clifford Berry. It was called the Atanasoff-Berry Computer (ABC). In 1943 an electronic computer name the Colossus was built for the military. Other developments continued in 1946 the first general– purpose digital computer, the Electronic Numerical Integrator and Computer (ENIAC) was built. It is said that this computer weighed 30 tons, and had 18,000 vacuum tubes which were used for processing. When this computer was turned on for the first time lights dim in sections of Philadelphia. Computers of this generation could only perform a single task, and they had no operating system.

1.3.2 Second Generation 1947 – 1962

This generation of computers used transistors instead of vacuum tubes which were more reliable. In 1951 the first computer for commercial use was introduced to the public; the Universal Automatic Computer (UNIVAC 1). In 1953 the International Business Machine (IBM) 650 and 700 series computers made their mark in the computer world. During this generation of computers over 100 computer programming languages were developed, computers had memory and operating systems. Storage media such as tape and disk were in use also were printers for output.

1.3.3 Third Generation: 1963 - Present

The invention of integrated circuit brought us the third generation of computers. With these invention computers became smaller, more powerful more reliable and they are able to run many different programs at the same time. In1980 Microsoft Disk Operating System (MS-Dos) was born and in 1981 IBM introduced the personal computer (PC) for home and office use. Three years later Apple gave us the Macintosh computer with its icon-driven interface and the 90s gave us Windows operating system.

As a result of the various improvements to the development of the computer, we have seen the computer being used in all areas of life. It is a very useful tool that will continue to experience new development as time passes.

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1.4 Classification of Computers

Computers differ based on their data processing abilities. They are classified according to purpose, data handling, and functionality.

1.4.1 Analog Computers

It accepts analog input and provides analog output information. It represents physical quantities like acceleration, temperature, distance or voltages in mechanical or electrical circuits and does not need any storage device. Examples of the analog computer are thermometer, speedometer and analog clock.

1.4.2 Digital Computer

This computer accepts digital input and provides digital output after processing information and the operation are in a binary system of 0 and 1. By manipulating the binary digits and numbers it can perform any task like analyze data, mathematical calculations etc. Examples of digital computers are Apple Macintosh, IBM PC.

1.4.3 Hybrid Computer

This computer is the combination of both analog and digital computers in terms of speed and accuracy. Hybrid computers can measure physical and digital quantities. Examples of the hybrid computer are the machine measure heartbeat in hospital, devices installed fuel pumps.

1.5 Types of Computer

There are many types of computers, some of which are given below:

1.5.1 Super Computer

The fastest and most powerful type of computer Supercomputers are very expensive and are employed for specialized applications that require immense amounts of mathematical calculations. For example, weather forecasting requires a supercomputer. Other uses of supercomputers include animated graphics, fluid dynamic calculations, nuclear energy research, and petroleum exploration.

The chief difference between a supercomputer and a mainframe is that a supercomputer channels all its power into executing a few programs as fast as possible, whereas a mainframe uses its power to execute many programs concurrently.

1.5.2 Mainframe Computer

A very large and expensive computer that is capable of supporting hundreds or even thousands of users simultaneously. In the hierarchy that starts with a simple microprocessor (in watches,

for example) at the bottom and moves to supercomputers at the top, mainframes are just below supercomputers. In some ways, mainframes are more powerful than supercomputers because they support more simultaneous programs. But supercomputers can execute a single program faster than a mainframe.

1.5.3 Mini Computer

A midsized computer called minicomputers lie between *workstations* and *mainframes*. In the past decade, the difference between large minicomputers and small mainframes has blurred, however, as has the distinction between small minicomputers and workstations. But in general, a minicomputer is a multiprocessing system capable of supporting from 4 to about 200 users simultaneously.

1.5.4 Micro Computer or Personal Computer

- **Desktop Computer:** a personal or micro-mini computer sufficient to fit on a desk.
- **Laptop Computer:** a portable computer complete with an integrated screen and keyboard. It is generally smaller in size than a desktop computer and larger than a notebook computer.
- **Laptop Computer/Digital Diary /Notebook /PDAs: a hand-sized computer.** Palmtops have no keyboard but the screen serves both as an input and output device. It is a terminal or desktop computer in a network. In this context, the workstation is just a generic term for a user's machine (client machine) in contrast to a "server" or "mainframe."

1.6 Components of a Computer

A computer system consists of both hardware and information stored on hardware. Information stored on computer hardware is often called software.

The hardware components of a computer system are the electronic and mechanical parts.

The software components of a computer system are the data and the computer programs.

1.6.1 Hardware

This refers to any part that is tangible (Touchable, Physical). The major hardware components of a computer system are:

- Processor
- Main memory
- Secondary memory

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- Input & Output devices
- Slots
- Ports
- Buses

These parts are typically accommodated within the laptop or desktop unit itself, except for the desktop keyboard and mouse. What is likely the most important piece of hardware is the microprocessor chip known as the central processing unit (CPU).

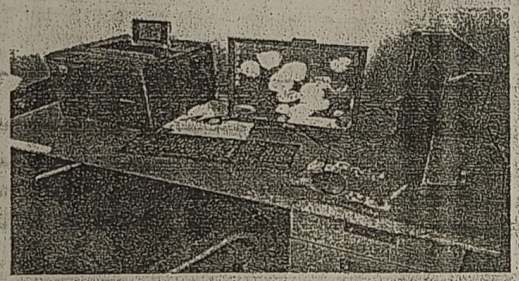

Figure 3: A typical installation of computer hardware

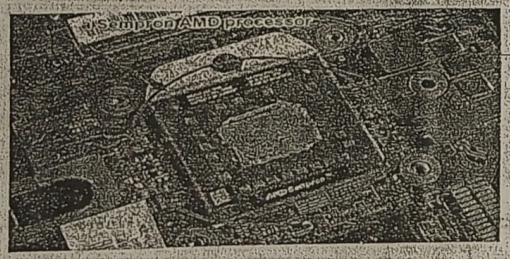

Figure 4: A Diagrammatic Representation of Mother Slot where CPU is fixed

Hard disk drive: HDDs are mechanical by design and store data on magnetic and metallic platters, which are read by the read/write

heads. This makes this type of drive rather changeable and likely to fail if used improperly.

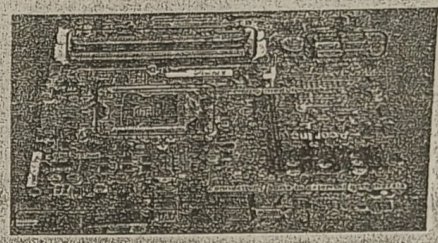

Figure 5: Memory Slots, Ports, and Internal Buses

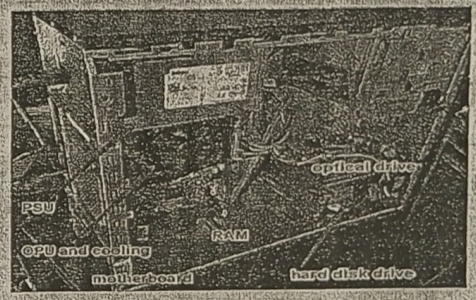

Figure 6: Different Parts of the System Unit

Connectivity to a motherboard can be internal or external. Internal devices that connect to the motherboard include:

- The microprocessor (CPU).
- Disk drive.
- Random access memory (memory modules).
- Power supply unit (PSU).

External peripherals that connect to the motherboard include:

- Monitor.

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- Keyboard.
- Mouse.
- Printer.

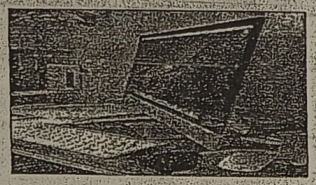

Figure 7: Different devices Hardware like Mouse, Keyboard, Monitor and cables

1.6.2 Computer Software

Software refers to the instructions, programs, data, and protocols which run on hardware. It can be divided into two major categories. The visible part, non-touchable part or programs in the computer system are called software. Like human brain thinking or instructions for the microprocessor is termed as software. The software is written in computer languages such as Visual Basic, C/C++, and Java.

Figure 8: A Diagram shows a visualized form of Software

i) System Software

This is also commonly known as an operating system (OS). The system manages other software and devices inside the computer.

In a typical setup, the operating system is like the motherboard software. It is the first thing that is installed, followed by applications and utility software. Three popular operating systems for traditional computers include Windows, Mac OS X, and Linux.

Popular mobile operating systems include Android OS, iPhone OS, Windows Phone OS, and Firefox OS.

ii) Application Software

This is designed for end users. This software is meant to perform a specialized assignment and output useful information.

An example would be a word processing application that one uses to compose a letter or a brochure, such as Microsoft Word.

a) General Purpose Application Software: A collection of application software can come in a package that is commonly known as a software suite or General Purpose Application Software. A typical suite includes software for word processing, presentations, graphic design, and spreadsheets. Examples include Microsoft Office, OpenOffice, and iWork, Dreamweaver9.0.

Special Purpose Application Software: This is designed for a specialized assignment and output useful information. Examples include Adobe Photoshop, Corel Draw, and AutoCAD.

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02

COMPUTER MEMORY

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COMPUTER MEMORY

2.1 What is Computer Memory?

The computer memory is a storage area inside the computer system. Memory is mainly divided into two types: Primary Memory and Secondary Memory.

Primary Memory - Primary memory is computer memory that a processor or computer accesses first or directly. It allows a processor to access running execution applications and services that are temporarily stored in a specific memory location. Primary memory is also known as primary storage or main memory.

Secondary Memory - Secondary memory is where programs and data are kept on a long-term basis. Common secondary storage devices are the hard disk and optical disks. The hard disk has enormous storage capacity compared to primary memory. The hard disk is usually contained inside the case of a computer.

2.1.1 Types of Primary Memory

1) RAM (Random Access Memory) - Random access memory (RAM) is a type of data storage used in computers that are generally located on the motherboard. This type of memory is volatile and all information that was stored in RAM is lost when the computer is turned off. There are two main types of RAM: dynamic random access memory (DRAM), or Dynamic RAM, and static random access memory (SRAM).

DRAM: The RAM in most personal computers (PC's) is Dynamic RAM. These RAM's are made of capacitors. All dynamic RAM chips have to need a permanent refresh every few milliseconds by rewriting the data to the module.

SRAM: Static RAM (SRAM) is a lot faster and does not require refreshing. It made from a complex circuitry system called a flip-flop. SRAM retains information and is able to operate at higher speeds than DRAM. It's common to see PC manufacturers use DRAM and also used in supercomputers.

2) ROM (Read Only Memory) - Read-only memory (ROM) is a type of storage medium that permanently stores data on personal

computers (PCs) and other electronic devices. It contains the programming needed to start a PC, which is essential for boot-up; it performs major input/output tasks and holds programs or software instructions. Because ROM is read-only, it cannot be changed; it is permanent and non-volatile, meaning it also holds its memory even when power is removed. There are numerous ROM chips located on the motherboard and a few on expansion boards. The chips are essential for the basic input/output system (BIOS), boot up, reading and writing to peripheral devices, basic data management, and the software for basic processes for certain utilities.

However, these types of non-volatile memory can be altered and are often referred to as programmable ROM. One of the original forms of non-volatile memory was mask-programmed ROM. It was designed for specific data such as bootstrap, which contains the startup code. Mask-programmed ROM can never be changed. Because ROM cannot be changed and is read-only, it is mainly used for the manufacturing company information and ownership purposes.

3) Cache Memory - Cache memory is a small-sized type of volatile computer memory that provides high-speed data access to a processor and stores frequently used computer programs, applications, and data. It is the fastest memory in a computer, and is typically integrated onto the motherboard and directly embedded in the processor or main random access memory (RAM). Cache memory provides faster data storage and access by storing instances of programs and data routinely accessed by the processor. Thus, when a processor requests data that already has an instance in the cache memory it does not need to go to the main memory or the hard disk to fetch the data.

Cache memory can be primary or secondary cache memory, with primary cache memory directly integrated into (or closest to) the processor. In addition to hardware-based cache, cache memory also can be a disk cache, where a reserved portion of disk stores and provides access to frequently accessed data/applications from the disk.

4) Register - A processor register (CPU register) is one of a small set of data holding places that are part of the computer processor. A register may hold an instruction, a storage address, or any kind of

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data (such as a bit sequence or individual characters). Some instructions specify registers as part of the instruction.

2.1.2 Types of Secondary Memory

1) Magnetic Tape: Magnetic tape is one of the oldest technologies for electronic data storage. The tape has largely been displaced as a primary and backup storage medium, but it remains well-suited for archiving because of its high capacity, low cost, and long durability. It is a linear recording system that is not good for random access. If the tape is part of a library, robotic selection and loading of the right cartridge into a tape drive adds more latency. In an archive, such latencies are not an issue. With tape archiving, there is no online copy for quick retrieval, as everything is vaulted for the long term.

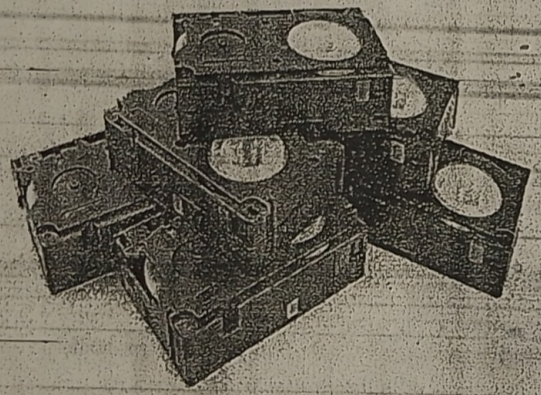

Figure 9: Typical Hard Disk

2) Hard – Disk: The hard disk was created in 1953 by engineers at IBM who wanted to find a way to provide random access to high capacities of data at a low cost. The disk drives developed were the size of refrigerators, could store 3.75 megabytes of data and began shipping in 1956. Memorex, Seagate, and Western Digital were other early vendors of hard disk drive technology. Hard disk drive form-factor size has continued to decrease as the technology evolves. By the mid-1980s, 3.5-inch and 2.5-inch form factors were

introduced, and it was at this time they first became a standard in personal computers (PCs).

Hard disk drive density has increased since the technology was first developed. The first hard disk drives were able to store megabytes of data, while today they are in the terabyte (TB) range. Hitachi released the first 1 TB hard drives in 2007. In 2017, HGST announced the first 10 TB hard drives.

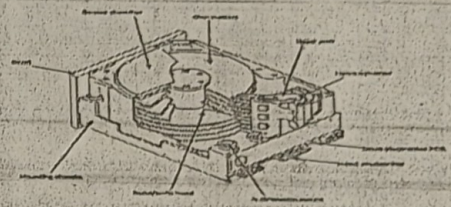

Figure 10: Parts of a Hard Disk

Most basic hard drives consist of a number of disk platters that are positioned around a spindle inside a sealed chamber. The chamber also includes read-and-write heads and motors. The motor is used to spin the platters, which hold the data, at up to 15,000 rotations per minute (a higher rpm number result in a good performance). As the platters spin, a second motor controls the position of the read-and-write heads that record information to, and read information from, tracks on each platter.

3) CD (Compact Disk) - A compact disc is a portable storage medium that can be used to record, store and playback audio, video and other data in digital form. A standard compact disc measure 4.7 inches, or 120 millimeters (mm), across, is 1.2 mm thick, weighs between 15 grams and 20 grams, and has a capacity of 80 minutes of audio, or 650 megabytes (MB) to 700 MB of data.

4) DVD (Digital Versatile Disk) - DVD is an optical disc technology with a 4.7-gigabyte storage capacity on a single-sided, one-layered disk, which is enough for a 133-minute movie. DVDs can be single- or double-sided, and can have two layers on each side; a double-sided, two-layered DVD will hold up to 17 gigabytes of

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video, audio, or other information. This compares to 650 megabytes (0.65 gigabytes) of storage for a CD-ROM disk.

DVD uses the MPEG-2 file and compression standard. MPEG-2 images have four times the resolution of MPEG-1 images and can be delivered at 60 interlaced fields per second where two fields constitute one image frame. (MPEG-1 can deliver 30 non-interlaced frames per second.) Audio quality on DVD is comparable to that of current audio compact discs.

Short for Digital Versatile Disc or Digital Video Disc, a DVD or DVD-ROM is a disc capable of storing large amounts of data on one disc the size of a standard Compact Disc. CD/DVD drives were first sold in 1997. They are widely used for storing and viewing movies and other data. To play DVDs on a computer, you must have a DVD drive and a software DVD player. The picture to the right is an example of what a DVD movie may look like, which in this example is a picture of the Matrix DVD movie.

5) **VCD (Video Compact Disk)** - Short for Video Compact Disc VCD is a CD (CD-R or CD-RW) that contains moving pictures and sound. A VCD can hold up to 80 minutes of motion video and stereo sound on a 700MB CD. VCDs use the MPEG-1 encoding standard. On a VCD you can have chapters, video, digital images, album slideshows, audio, or a combination of media files. A VCD can be played on most DVD players and on computer DVD-ROMs and computer CD-ROMs with the aid of software.

6) Flash Memory - Flash memory is a type of nonvolatile memory that erases data in units called blocks. A block stored on a flash memory chip must be erased before data can be written, or programmed, to the microchip. Flash memory retains data for an extended period of time whether a flash-equipped device is powered on or off.

Dr. Fujio Masuoka is credited with the invention of flash memory when he worked for Toshiba in the 1980s. Masuoka's colleague, Shoji Ariizumi, coined the term flash because the process of erasing all the data from a semiconductor chip reminded him of the flash of a camera.

2.2 Memory Hierarchy

In computer architecture, the memory hierarchy separates computer storage into a hierarchy based on response time. Since response time, complexity, and capacity are

related, the levels may also be distinguished by their performance and control technologies.[1] Memory hierarchy affects performance in computer architectural design, algorithm predictions, and lower level programming constructs involving locality of reference.

Designing for high performance requires considering the restrictions of the memory hierarchy, i.e. the size and capabilities of each component. Each of the various components can be viewed as part of a hierarchy of memories in which each member mi is typically smaller and faster than the next highest member of the hierarchy. To limit waiting by higher levels, a lower level will respond by filling a buffer and then signaling to activate the transfer.

There are four major storage levels

1. Internal – Processor registers and cache.
2. Main – the system RAM and controller cards.
3. On-line mass storage – Secondary storage.
4. Off-line bulk storage – Tertiary and Off-line storage.

This is a general memory hierarchy structuring. Many other structures are useful. For example, a paging algorithm may be considered as a level for virtual memory when designing computer architecture and one can include a level of near-line storage between online and offline storage.

4bit = 1 Nibble	8bit = 1 Byte
16 Bit = 1 Word	32 Bit = Double Word
64 Bit = Quad Word	1024 Byte = 1KB
1024KB = 1MB	1024MB=1GD
1024 GB = 1TB	

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*KB means Kilo Byte, MB Means Mega Byte, GB Means Giga Byte and TB Means Tera Byte

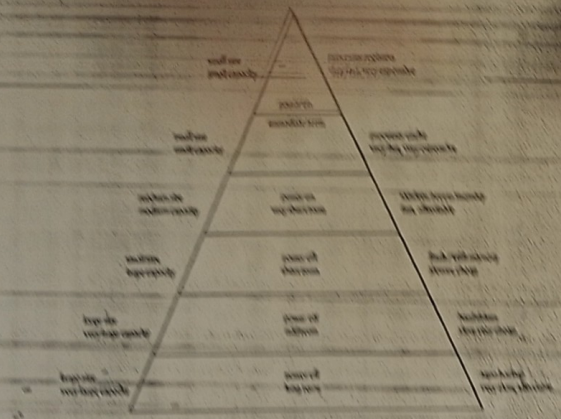

Figure 11: Computer Memory Hierarchy

03

CENTRAL PROCESSING UNIT OR PROCESSOR OR MICRO PROCESSOR

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CENTRAL PROCESSING UNIT (CPU)/PROCESSOR OR MICRO-PROCESSOR

3.1 Central Processing Unit (CPU)/Processor or Micro-Processor

A central processing unit (CPU) is the electronic circuitry within a computer that carries out the instructions of a computer program by performing the basic arithmetic, logical, control and input/output (I/O) operations specified by the instructions.

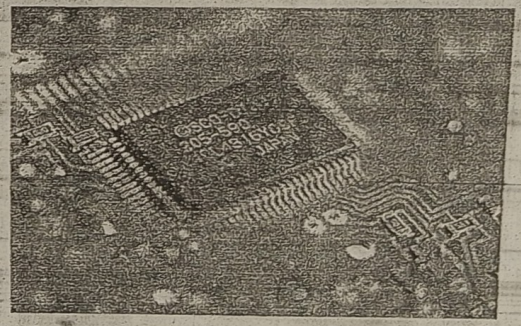

Figure 12: A Microprocessor Chip Fixed on Motherboard

A processor, or "microprocessor," is a small chip that resides in computers and other electronic devices. Its basic job is to receive input and provide the appropriate output. ... The central processor of a computer is also known as the CPU, or "central processing unit."

The main components of CPU help it in performing various functions. The three components of the CPU are following:

1. Arithmetic Logic Unit
2. Control Unit
3. Registers

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Each of the components is explaining one-by-one under the following headings:-

1) Arithmetic Logic Unit (ALU): There is electronic circuitry in the arithmetic logic unit which executes all arithmetic and logical operations. Its function is obvious from its name. It performs arithmetic calculations like as addition, subtraction, multiplication, and division as well as comparisons. The unit can compare numbers, letters, or special characters. There can be more than one Arithmetic logic unit in a CPU, and these ALUs can also be used for the purpose of maintaining timers that help run the computer.

2) Control Unit (CU): There is circuitry in the control unit which uses electrical signals to instruct the whole computer system for carrying out or executing, already stored program instructions. Its name clearly shows that it controls and coordinates computer components. It extracts instructions from memory and decodes and executes them. In fact, it regulates the flow of information through the processor. In short, it can be said, this component receives, decodes, stores results and manages execution of data that flows through the CPU. Its communication with both arithmetic unit and memory is inevitable.

3) Registers: Registers are temporary storage areas which are responsible for holding the data that is to be processed. They store the instructions and data in a processor. This data is further used by Control Unit. There are some registers that are set aside for specific tasks; these generally include a program counter, stack, and flags.

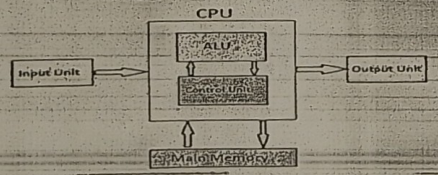

Figure 13: Block Diagram of CPU

3.2 Computer Bus

A bus, in computing, is a set of physical connections (cables, printed circuits, etc.) which can be shared by multiple hardware components in order to communicate with one another.

The purpose of buses is to reduce the number of "pathways" needed for communication between the components, by carrying out all communications over a single data channel. This is why the metaphor of a "data highway" is sometimes used. In reality, each bus is generally constituted of 50 to 100 distinct physical lines, divided into three subassemblies:

Address Bus: The address bus (sometimes called the memory bus) transports memory addresses which the processor wants to access in order to read or write data. It is a unidirectional bus.

Date Bus: The data bus transfers instructions coming from or going to the processor. It is a bidirectional bus.

Control Bus: The control bus (or command bus) transports orders and synchronization signals coming from the control unit and traveling to all other hardware components. It is a bidirectional bus, as it also transmits response signals from the hardware.

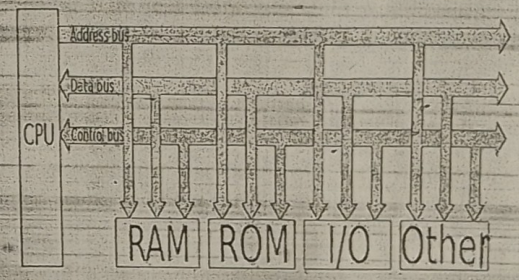

Figure 14: A System Bus

3.3 Computer Port

When referring to a physical device, a hardware port or peripheral port is a hole or connection found on the front or back of a computer. Ports allow computers to access external devices such as printers. Below is a short listing of the different computer ports you may find on a computer. The picture shows an example of a type of port on the back of a computer.

A computer port is also called as a Communication Port as it is responsible for communication between the computer and its

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peripheral device. Generally, the female end of the connector is referred to as a port and it usually sits on the motherboard.

In Computers, communication ports can be divided into two types based on the type of protocol used for communication. They are Serial Ports and Parallel Ports.

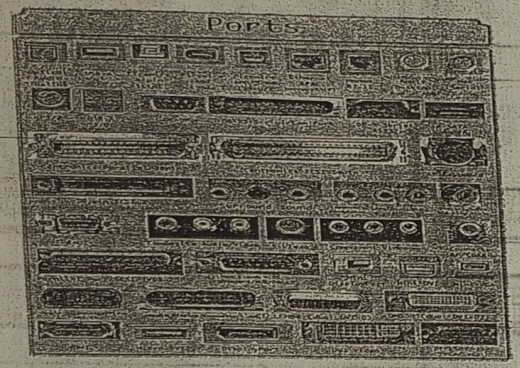

Figure 15: Different Ports in Computer System

A serial port is an interface through which peripherals can be connected using a serial protocol which involves the transmission of data one bit at a time over a single communication line. The most common type of serial port is a D-Subminiature or a D-sub connector that carry RS-232 signals.

A parallel port, on the other hand, is an interface through which the communication between a computer and its peripheral device is in a parallel manner i.e. data is transferred in or out in parallel using more than one communication line or wire. The printer port is an example of the parallel port.

3.4 Computer language

A programming language is a formal language that specifies a set of instructions that can be used to produce various kinds of output. Programming languages generally consist of instructions for a computer. Programming languages can be used to create programs that implement specific algorithms.

A language is the main medium of communicating between the Computer systems and the most common are the programming languages. As we know a Computer only understands binary numbers that is 0 and 1 to perform various operations but the languages are developed for different types of work on a Computer. A language consists of all the instructions to make a request to the system for processing a task. From the first generation and now the fourth generation of the Computers there were several programming languages used to communicate with the Computer. Here we will go into the detail of the Computer language and its types.

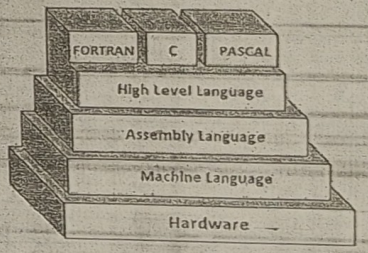

Figure 16: Computer Languages and its types

A Computer language includes various languages that are used to communicate with a Computer machine. Some of the languages like programming language which is a set of codes or instructions used for communicating the machine. Machine code is also considered as a computer language that can be used for programming. And HTML also is a computer language or a markup language but not a programming language. Similarly, there are different types of languages developed for different types of work to be performed by communicating with the machine. But all the languages that are now available are categorized into two basic types of languages including Low-level language and High-level language.

3.4.1 High-Level Language (HLL)

The high-level languages are the most used and also more considered programming languages that help a programmer to read, write and maintain. It is also the third generation language that is used and also running till now by many programmers. They are less

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independent of a particular type of Computer and also require a translator that can convert the high-level language to machine language. The translator may be an interpreter and Compiler that helps to convert into binary code for a Computer to understand. There are various high-level programming languages like C, FORTRAN or Pascal that are less independent and also enable the programmer to write a program.

The Compiler plays an important role on the Computer as it can convert to machine language and also checks for errors if any before executing. There are several high-level languages that were used earlier and also now like COBOL, FORTRAN, BASIC, C, C++, PASCAL, LISP, Ada, Algol, Prolog, and Java. It is user-friendly as the programs are written in English using words, symbols, characters, numbers that need to be converted to machine code for processing.

3.4.2 Low-Level Language (LLL)

Low-level languages are the machine codes in which the instructions are given in machine language in the form of 0 and 1 to a Computer system. It is mainly designed to operate and handle all the hardware and instructions set architecture of a Computer.

Machine Language is one of the low-level programming languages which is the first generation language developed for communicating with a Computer. It is written in machine code which represents 0 and 1 binary digits inside the Computer string which makes it easy to understand and perform the operations. As we know a Computer system can recognize electric signals so here 0 stands for turning off electric pulse and 1 stands for turning on the electric pulse. It is very easy to understand by the Computer and also increases the processing speed.

The main advantage of using Machine language is that there is no need for a translator or interpreter to translate the code, as the Computer directly can understand. But there are some disadvantages also like you have to remember the operation codes, memory address every time you write a program and also hard to find errors in a written program. It is a machine dependent and can be used by a single type of Computer.

3.4.3 Assembly Language (Intermediate Level Languages)

Assembly Language is the second generation programming language that has almost similar structure and set of commands as

Machine language. Instead of using numbers like in Machine languages here we use words or names in English forms and also symbols. The programs that have been written using words, names and symbols in assembly language are converted to machine language using an Assembler. Because a Computer only understands machine code languages that's why we need an Assembler that can convert the Assembly level language to Machine language so the Computer gets the instruction and responds quickly.

The main disadvantage of this language is that it is written only for a single type of CPU and does not run on any other CPU. But its speed makes it the most used low-level language till today which is used by many programmers.

3.5 Translator in Computer

A program written in the high-level language is called as source code. To convert the source code into machine code, translators are needed.

A translator takes a program written in source language as input and converts it into a program in the target language as output.

It also detects and reports the error during translation. It translates the high-level language program input into an equivalent machine language program output. It also provides diagnostic messages wherever the programmer violates specification of the high-level.

Different type of translators

The different types of the translator are as follows:

3.5.1 Compiler

The compiler is a translator which is used to convert programs in a high-level language to low-level language.

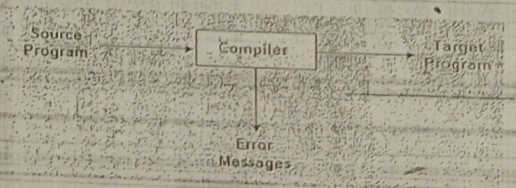

Figure 17: Function of Compiler

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It translates the entire program and also reports the errors in source program encountered during the translation.

3.5.2 Interpreter

The interpreter is a translator which is used to convert programs in a high-level language to low-level language. The interpreter translates line by line and reports the error once it encountered during the translation process. It directly executes the operations specified in the source program when the input is given by the user. It gives better error diagnostics than a compiler.

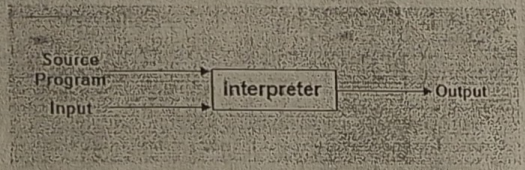

Figure 18: Function of Interpreter

3.5.3 Assembler

Assembler is a translator which is used to translate the assembly language code into machine language code.

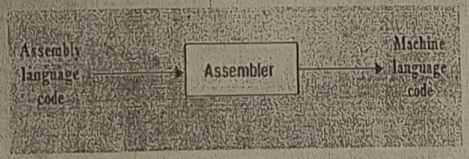

Figure 19: Function of Assembler

3.6 DIFFERENCES BETWEEN COMPILER AND INTERPRETER

S/No	Compiler	Interpreter
1	Performs the translation of a program as a whole.	Performs statement by statement translation.
2	Execution is faster.	Execution is slower.
3	Requires more memory as linking is needed for the generated intermediate object code.	Memory usage is efficient as no intermediate object code is generated.
4	Debugging is hard as the error messages are generated after scanning the entire program only.	It stops translation when the first error is met. Hence, debugging is easy.
5	Programming languages like C, C++ uses compilers.	Programming languages like Python, BASIC, and Ruby uses interpreters.

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